VIT-AP UNIVERSITY, ANDHRA PRADESH

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech **Date:**
Faculty Name: Prof. S.Gopikrishnan **School:** SCOPE
Student name: **Reg. no.:**

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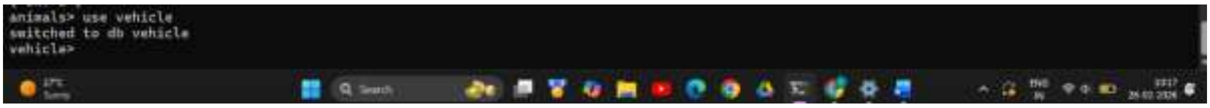
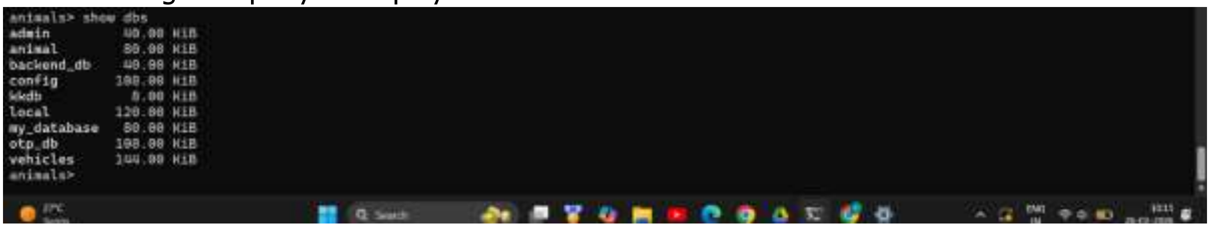
1. Use MongoDB to implement the following DB operations

1. Create a database called 'vehicles' and *write* a MongoDB query to select database as "vehicles".



```
animals> use vehicle
switched to db vehicle
vehicle>
```

2. Write a MongoDB query to display all the databases



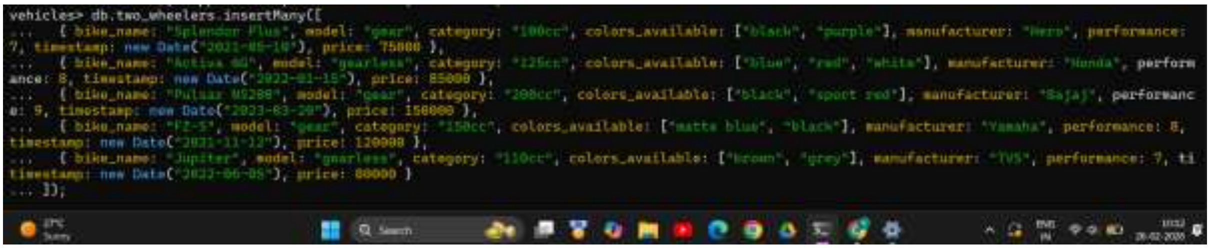
```
animals> show dbs
admin          40.00 KiB
animal         80.00 KiB
backend_db     40.00 KiB
config        108.00 KiB
Wkdb            8.00 KiB
local         120.00 KiB
my_database    80.00 KiB
otp_db        108.00 KiB
vehicles      144.00 KiB
animals>
```

.

3. Create a collection called 'two_wheelers'. (use capping) and Create a collection called 'four_wheelers'.

```
vehicles> // Capped collection (size in bytes)
... db.createCollection("two_wheelers", { capped: true, size: 5242880, max: 5000 });
...
... // Standard collection
... db.createCollection("four_wheelers");
```

4. Add 5 two-wheeler details to the collection named 'two_wheelers'. Each document consists of following fields as bike_name, model (gear or gearless), category (100cc, 125cc, 150cc, 200cc), colors_available (red, black, blue, sport red etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.



```
vehicles> db.two_wheelers.insertMany([
...   { bike_name: "Splendor Plus", model: "gear", category: "100cc", colors_available: ["black", "purple"], manufacturer: "Hero", performance: 7, timestamp: new Date("2021-05-10"), price: 75000 },
...   { bike_name: "Activa 6G", model: "gearless", category: "125cc", colors_available: ["blue", "red", "white"], manufacturer: "Honda", performance: 8, timestamp: new Date("2022-01-15"), price: 85000 },
...   { bike_name: "Pulsar NS200", model: "gear", category: "200cc", colors_available: ["black", "sport red"], manufacturer: "Bajaj", performance: 9, timestamp: new Date("2023-03-20"), price: 150000 },
...   { bike_name: "FZ-S", model: "gear", category: "150cc", colors_available: ["matte blue", "black"], manufacturer: "Yamaha", performance: 8, timestamp: new Date("2021-11-12"), price: 120000 },
...   { bike_name: "Jupiter", model: "gearless", category: "110cc", colors_available: ["brown", "grey"], manufacturer: "TVS", performance: 7, ti timestamp: new Date("2022-06-05"), price: 80000 }
... ]);
```

5. Add 5 four-wheeler details to the collection named 'four_wheelers'. Each document consists of following fields as vehicle_name, model (commercial or own), category (car, lorry, bus, mini truck, heavy truck, containers), variants (vxi, zxi, petrol, diesel

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etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.

```
}
vehicles> db.four_wheelers.insertMany([
...   { vehicle_name: "Swift", model: "own", category: "car", variants: ["vxi", "zxi", "pe
timestamp: new Date("2022-04-10"), price: 700000 },
...   { vehicle_name: "Eicher Pro", model: "commercial", category: "heavy truck", variants
timestamp: new Date("2020-08-22"), price: 2500000 },
...   { vehicle_name: "Fortuner", model: "own", category: "car", variants: ["diesel", "4x4
 new Date("2023-01-05"), price: 3500000 },
...   { vehicle_name: "Ace Gold", model: "commercial", category: "mini truck", variants: [
, timestamp: new Date("2021-09-15"), price: 500000 },
...   { vehicle_name: "City", model: "own", category: "car", variants: ["vxi", "petrol"],
Date("2022-12-10"), price: 1200000 }
... ]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('699fcf5b9c35c57c29cebeb3'),
    '1': ObjectId('699fcf5b9c35c57c29cebeb4'),
    '2': ObjectId('699fcf5b9c35c57c29cebeb5'),
    '3': ObjectId('699fcf5b9c35c57c29cebeb6'),
    '4': ObjectId('699fcf5b9c35c57c29cebeb7')
```

6. Write a MongoDB query to display all documents available in two_wheelers and four_wheelers.

```
    vehicle_name: 'Ashok Leyland Dost',
    model: 'commercial',
    category: 'mini truck',
    variants: [ 'diesel' ],
    manufacturer: 'Ashok Leyland',
    performance: 7,
    timestamp: ISODate('2023-04-01T00:00:00.000Z'),
    price: 900000
  },
  {
    _id: ObjectId('699fc66553494fc225cebead'),
    vehicle_name: 'Hyundai i20',
    model: 'own',
    category: 'car',
    variants: [ 'petrol', 'diesel' ],
    manufacturer: 'Hyundai',
    performance: 9,
    timestamp: ISODate('2023-06-25T00:00:00.000Z'),
    price: 850000
  },
  {
    _id: ObjectId('699fcf5b9c35c57c29cebeb3'),
    vehicle_name: 'Swift',
    model: 'own',
    category: 'car',
    variants: [ 'vxi', 'zxi', 'petrol' ],
    manufacturer: 'Maruti',
    performance: 8,
    timestamp: ISODate('2022-04-10T00:00:00.000Z'),
    price: 700000
  },
  {
    _id: ObjectId('699fcf5b9c35c57c29cebeb4'),
    vehicle_name: 'Eicher Pro',
    model: 'commercial',
    category: 'heavy truck',
    variants: [ 'diesel' ],
    manufacturer: 'Eicher',
```

VIT-AP UNIVERSITY, ANDHRA PRADESH

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech **Date:**
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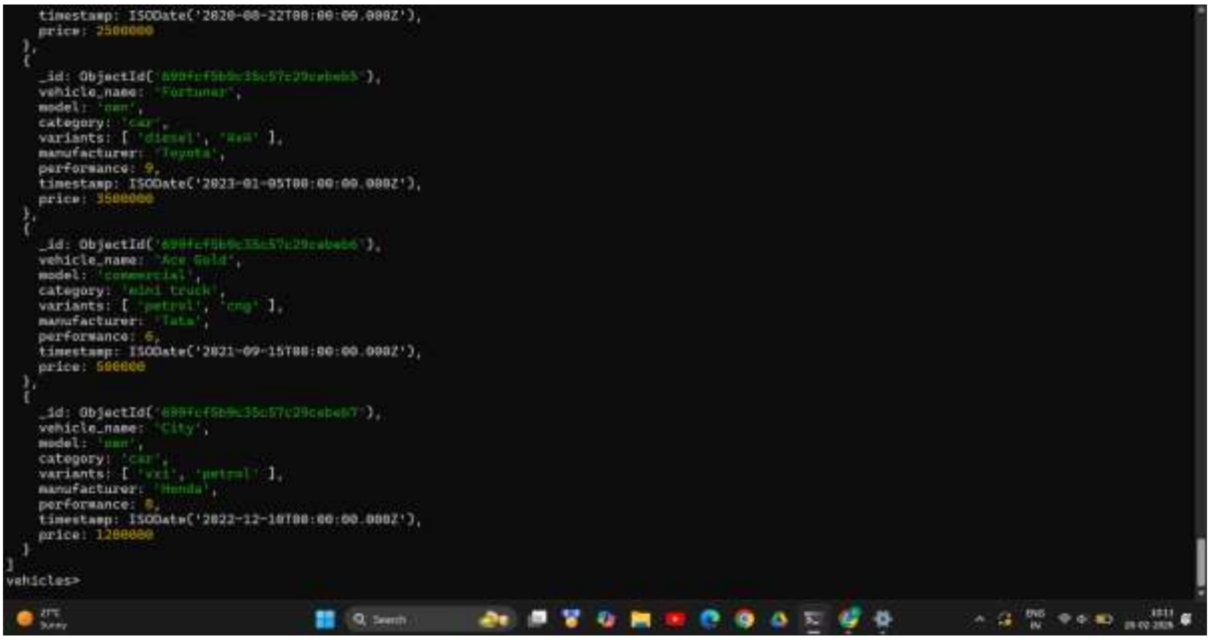
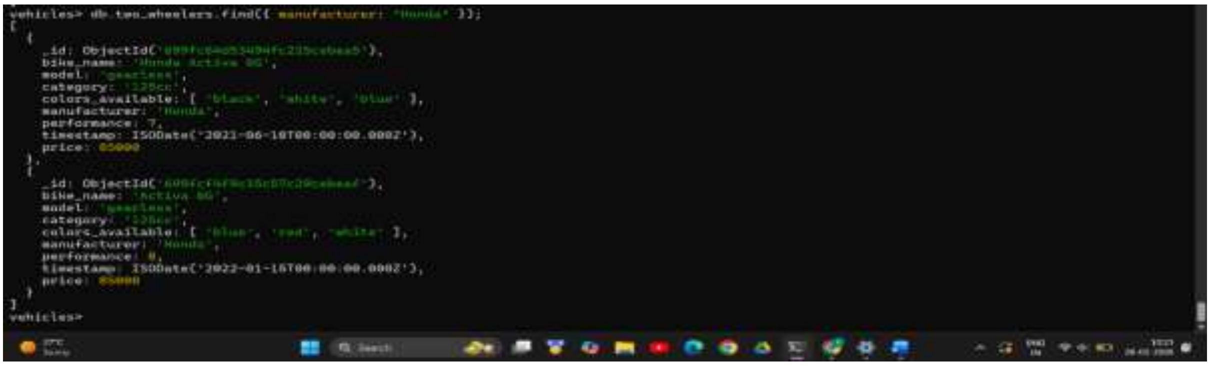


```
    timestamp: ISODate('2020-08-22T00:00:00.000Z'),
    price: 2500000
  },
  {
    _id: ObjectId('699fcf5b9c35c57c29cebeb5'),
    vehicle_name: 'Fortuner',
    model: 'oen',
    category: 'car',
    variants: [ 'diesel', '4x4' ],
    manufacturer: 'Toyota',
    performance: 9,
    timestamp: ISODate('2023-01-05T00:00:00.000Z'),
    price: 3500000
  },
  {
    _id: ObjectId('699fcf5b9c35c57c29cebeb6'),
    vehicle_name: 'Ace Gold',
    model: 'commercial',
    category: 'mini truck',
    variants: [ 'petrol', 'cng' ],
    manufacturer: 'Tata',
    performance: 6,
    timestamp: ISODate('2021-09-15T00:00:00.000Z'),
    price: 500000
  },
  {
    _id: ObjectId('699fcf5b9c35c57c29cebeb7'),
    vehicle_name: 'City',
    model: 'oen',
    category: 'car',
    variants: [ 'vxi', 'petrol' ],
    manufacturer: 'Honda',
    performance: 8,
    timestamp: ISODate('2022-12-10T00:00:00.000Z'),
    price: 1200000
  }
]
vehicles>
```

7. Write a MongoDB query to display only vehicle name and price in all the collection of the database

```
vehicles> db.two_wheelers.find({}, { bike_name: 1, price: 1, _id: 0 });
... db.four_wheelers.find({}, { vehicle_name: 1, price: 1, _id: 0 });
[
  { vehicle_name: 'Maruti Swift', price: 700000 },
  { vehicle_name: 'Tata Intra', price: 800000 },
  { vehicle_name: 'Mahindra Bolero', price: 950000 },
  { vehicle_name: 'Ashok Leyland Dost', price: 900000 },
  { vehicle_name: 'Hyundai i20', price: 850000 },
  { vehicle_name: 'Swift', price: 700000 },
  { vehicle_name: 'Eicher Pro', price: 2500000 },
  { vehicle_name: 'Fortuner', price: 3500000 },
  { vehicle_name: 'Ace Gold', price: 500000 },
  { vehicle_name: 'City', price: 1200000 }
]
vehicles>
```

8. Write a MongoDB query to display two_wheelers from a particular company



```
vehicles> db.two_wheelers.find({ manufacturer: 'Honda' });
[
  {
    _id: ObjectId('699fc0405340fc235cebea5'),
    bike_name: 'Honda Activa 6G',
    model: 'gearless',
    category: '125cc',
    colors_available: [ 'black', 'white', 'blue' ],
    manufacturer: 'Honda',
    performance: 7,
    timestamp: ISODate('2021-06-10T00:00:00.000Z'),
    price: 85000
  },
  {
    _id: ObjectId('699fcf9f9c35c57c29cebead'),
    bike_name: 'Activa 6G',
    model: 'gearless',
    category: '125cc',
    colors_available: [ 'blue', 'red', 'white' ],
    manufacturer: 'Honda',
    performance: 8,
    timestamp: ISODate('2022-01-15T00:00:00.000Z'),
    price: 85000
  }
]
vehicles>
```

VIT-AP UNIVERSITY, ANDHRA PRADESH

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech **Date:**
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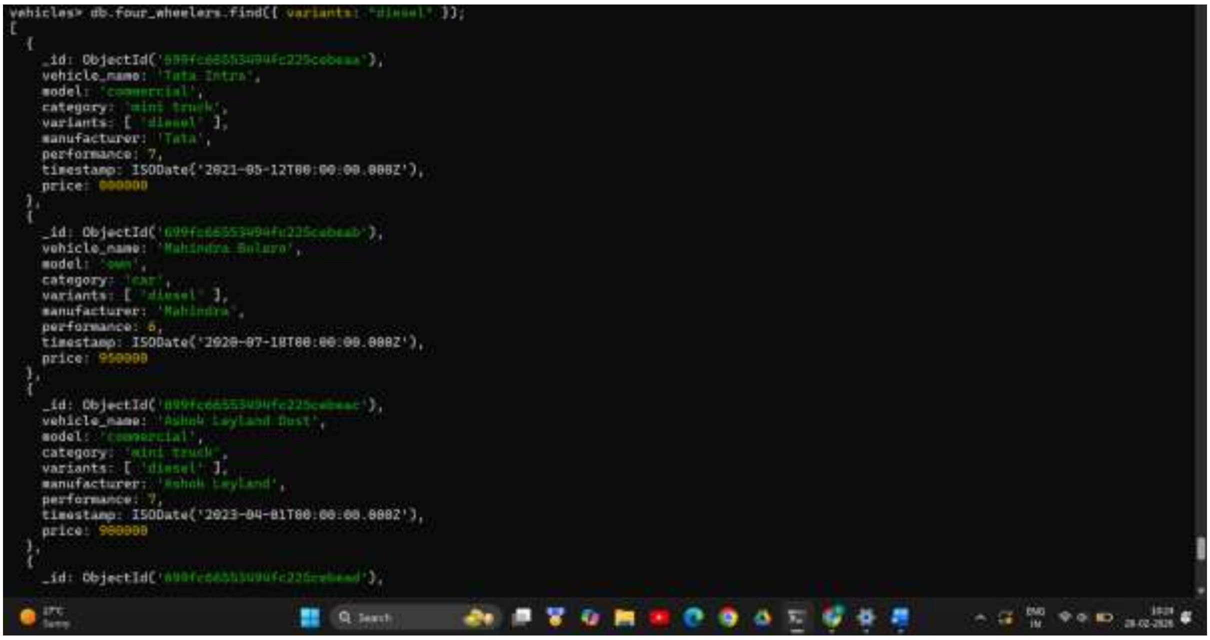
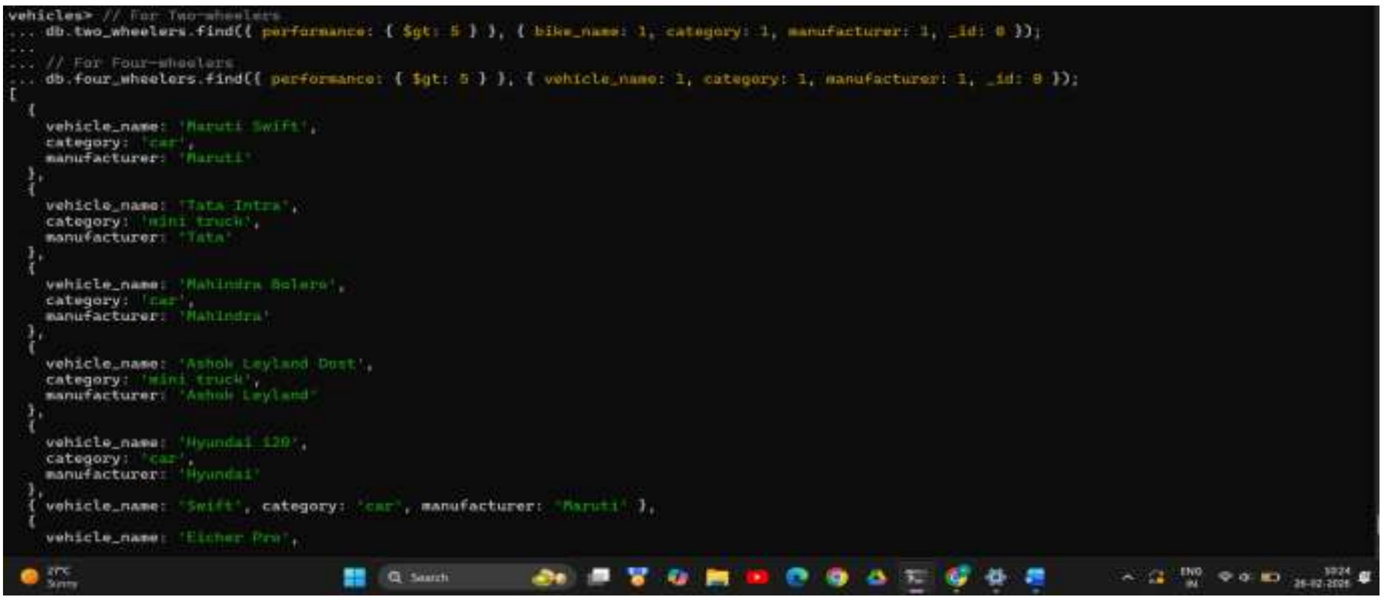
9. Write a MongoDB query to display four_wheelers available in diesel variants



10. Write a MongoDB query to display vehicles name, category and manufacturer details whose rating is more than 5.



VIT-AP UNIVERSITY, ANDHRA PRADESH

Lab Sheet 6: MongoDB Basic commands

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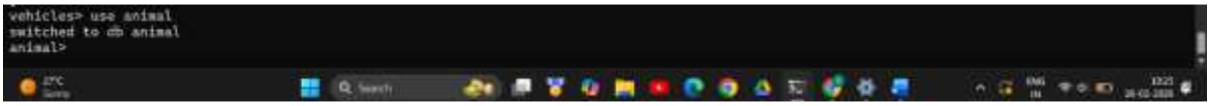
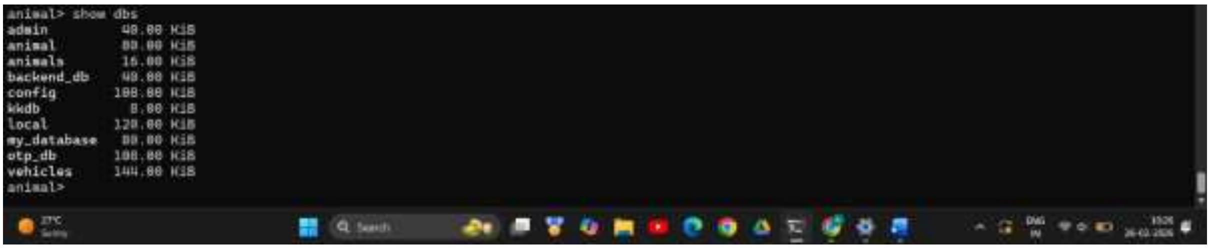
2. Use MongoDB to implement the following DB operations for a Zoo

1. Create a database called 'animal' and *write* a MongoDB query to select database as 'animal'.



2. Write a MongoDB query to display all the databases.



3. Create a collection called 'wild_animals'.(use capping) and Create a collection called 'domestic_animals'.

4. Add 5 wild_animal details to the collection named 'wild_animals'. Each document consists of following fields as animal_name, nature (harm or harmless),

VIT-AP UNIVERSITY, ANDHRA PRADESH

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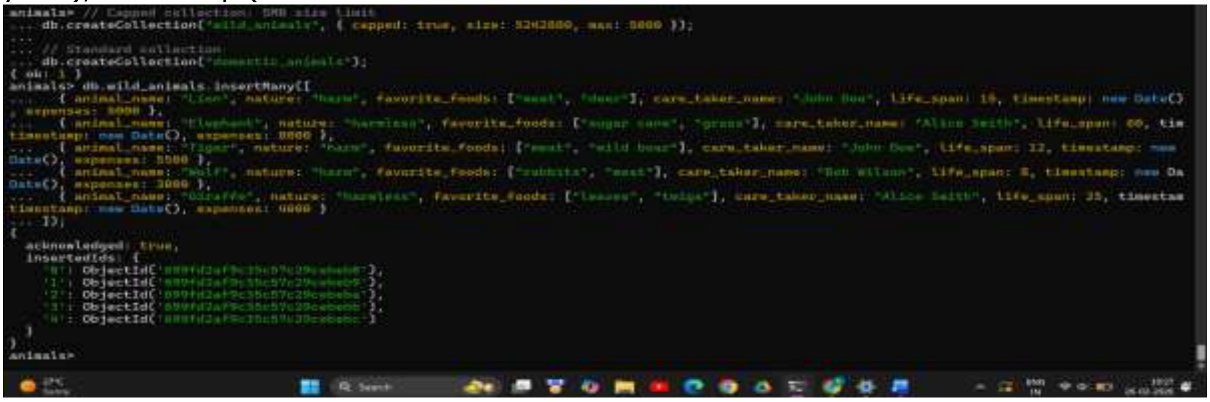
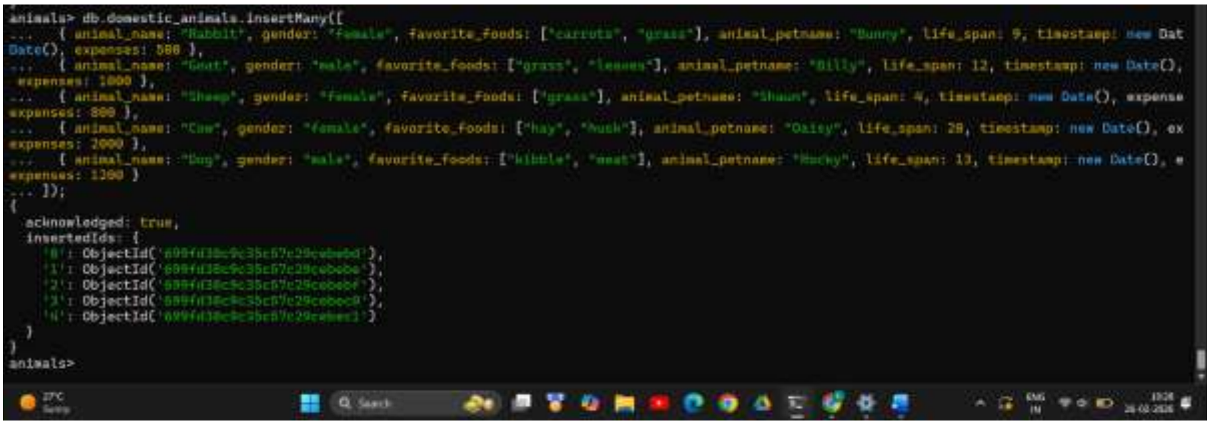
favorite_foods (meat, rabbits, deer etc) as array, care_taker_name, life span (in years), timestamp (when the anim



al registered at the Zoo) and expenses.

5. Add 5 domestic-animal details to the collection named ‘domestic_animals’. Each document consists of following fields as animal_name, gender (male or female), favorite_foods (meat, rabbits, deer etc) as array, animal_petname, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.



6. Write a MongoDB query to display all documents available in wild_animals and domastic_animals.

VIT-AP UNIVERSITY, ANDHRA PRADESH

Lab Sheet 6: MongoDB Basic commands

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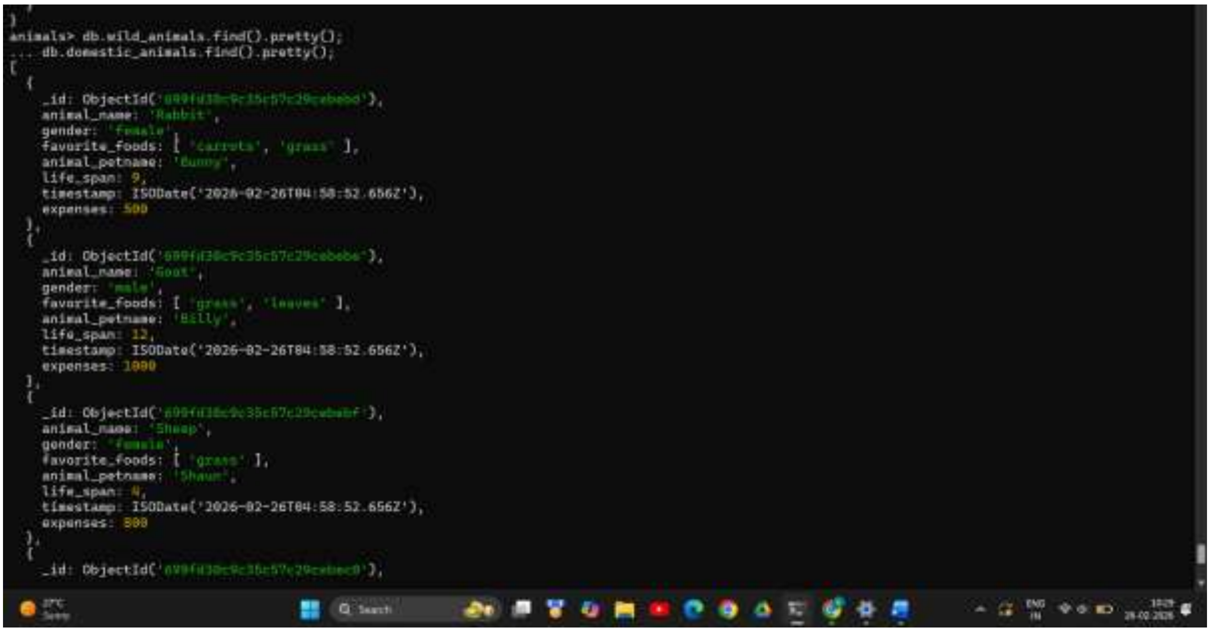
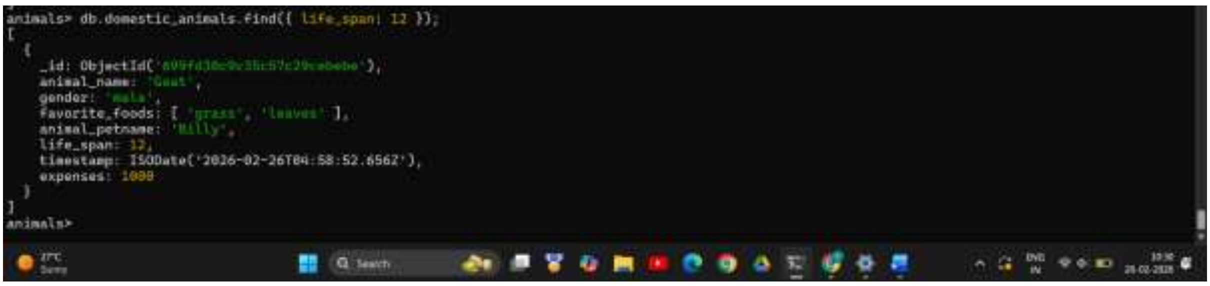


7. Write a MongoDB query to display only animal name and expenses in all the collection of the database

8. Write a MongoDB query to display domestic_animals whose life is a particular year



9. Write a MongoDB query to display wild_animals available under a particular care_taker

VIT-AP UNIVERSITY, ANDHRA PRADESH

Lab Sheet 6: MongoDB Basic commands

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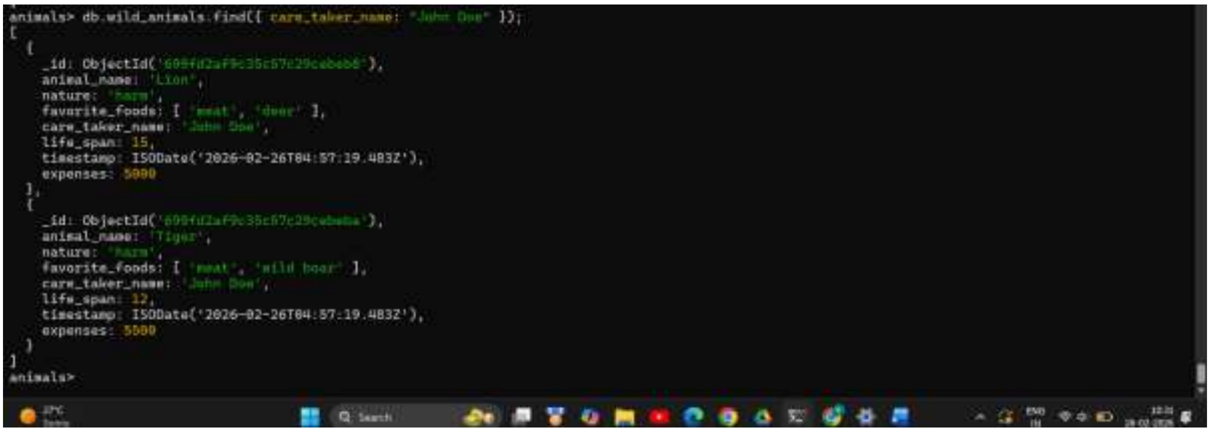
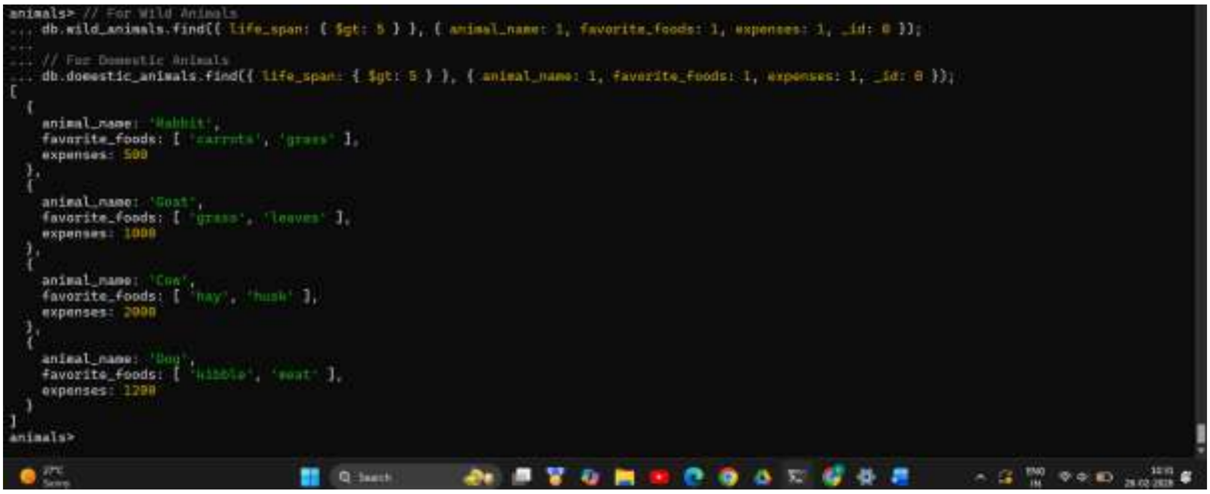


```
animals> db.wild_animals.find({ care_taker_name: "John Doe" });
[
  {
    _id: ObjectId('699fd2af9c35c57c29cebeb8'),
    animal_name: 'Lion',
    nature: 'harm',
    favorite_foods: [ 'meat', 'deer' ],
    care_taker_name: 'John Doe',
    life_span: 15,
    timestamp: ISODate('2026-02-26T04:57:19.483Z'),
    expenses: 5000
  },
  {
    _id: ObjectId('699fd2af9c35c57c29cebeba'),
    animal_name: 'Tiger',
    nature: 'harm',
    favorite_foods: [ 'meat', 'wild boar' ],
    care_taker_name: 'John Doe',
    life_span: 12,
    timestamp: ISODate('2026-02-26T04:57:19.483Z'),
    expenses: 5500
  }
]
animals>
```

10. Write a MongoDB query to display animal name, favorite_foods and expenses details whose lifespan is more than 5 years.



```
animals> // For Wild Animals
... db.wild_animals.find({ life_span: { $gt: 5 } }, { animal_name: 1, favorite_foods: 1, expenses: 1, _id: 0 });
...
... // For Domestic Animals
... db.domestic_animals.find({ life_span: { $gt: 5 } }, { animal_name: 1, favorite_foods: 1, expenses: 1, _id: 0 });
[
  {
    animal_name: 'Rabbit',
    favorite_foods: [ 'carrots', 'grass' ],
    expenses: 500
  },
  {
    animal_name: 'Goat',
    favorite_foods: [ 'grass', 'leaves' ],
    expenses: 1000
  },
  {
    animal_name: 'Cow',
    favorite_foods: [ 'hay', 'husk' ],
    expenses: 2000
  },
  {
    animal_name: 'Dog',
    favorite_foods: [ 'kibble', 'meat' ],
    expenses: 1200
  }
]
animals>
```